UNIT – III

Time Study • Rating • Allowances • Standard Time • PMTS

Syllabus (as given)

Time study – definition, time study equipment, selection of job, steps in time study. Breaking jobs into elements, recording information. Rating: systems of rating, standard rating, standard performance, scales of rating. Allowances: standard time determination, predetermined motion time study (PMTS), factors affecting rate of working, problems on allowances.

1. Time Study – Definition

Definition

Time study is a work-measurement technique for recording the times and rates of working for the elements of a specified job carried out under specified conditions, and for analysing the data so as to obtain the time necessary to carry out the job at a defined level of performance.

- Developed by F.W. **Taylor** (1881, Midvale Steel); the stop-watch is the basic tool, hence also called "stop-watch time study".
- Result: **Standard time** = the time a qualified, trained worker needs to do the job at standard performance, working in the specified method, including allowances for fatigue, personal needs and delays.

Objectives / uses

- To fix standard time for (i) production planning & scheduling, (ii) labour cost & product costing, (iii) wage-incentive plans, (iv) manpower & machine requirements, (v) assembly-line balancing, (vi) comparing methods, (vii) tendering / delivery dates.

Requirements for a good time study

- The method must be **standardised** first (method study before work measurement). The operator should be **qualified,** properly trained and willing; work-place conditions should be normal.
- Suitable for repetitive, short-cycle, manual work with enough volume to justify the cost; the work-study man must be trained and impartial; workers / union informed.

2. Time Study Equipment

1. **Stop-watch:** (a) ordinary (decimal-minute: 1 division = 0.01 min; 100 divisions per rev.); (b) split-hand / fly-back type; (c) decimal-hour watch (1 division = 0.0001 h); (d) electronic digital stopwatch / data recorder (0.01 s resolution, memory, can download).
2. **Time study board:** light plywood / plastic with a clip, holds the observation sheet and mounts the stop-watch (at elbow height).
3. **Time study observation sheets / forms:** for general information, element description, readings, rating, calculations.
4. Other: stop-clock, **video camera** / movie camera (for micromotion), tachometer / spring balance (to measure speed, force), steel tape, pencils, calculator / laptop, **Time-study software**, stopwatch counters.



crown (start / stop / reset)

Decimal-minute stop-watch

- 100 divisions = 1 min (1 div = 0.01 min)
- small dial counts 30 min
- fly-back: press = return to zero & restart
- mounted on the board, held at elbow level

Fig 3.1 Decimal-minute stop-watch (sketch)

Note: A watch should be sturdy, accurate, checked / calibrated periodically; electronic timers are more accurate and give less reading error.

3. Selection of the Job

- Reasons for study: new job / new product; change of method, material or tools; complaints about the standard (workers or union); a bottleneck or high-cost operation; poor machine utilisation; a basis for incentives.
- Selection by: **economic** importance (high labour content, large volumes), **technical** (quality, repetitiveness), **human** factors (fatigue, complaint). Choose a **qualified & average operator** (neither the fastest nor slowest).
- Inform the supervisor, union representative and operator; explain the purpose; stand where you can see the work without disturbing the operator.

4. Steps in Time Study (ILO – 8 steps)

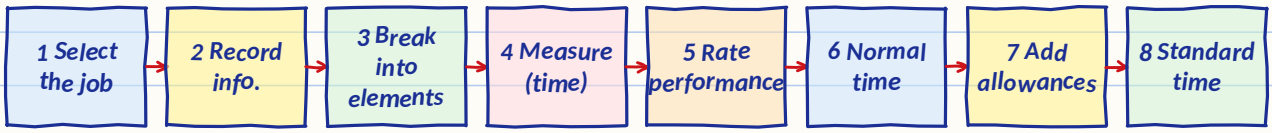


Fig 3.2 Steps in a stop-watch time study

1. **Select** the job to be studied (and the operator).
2. **Obtain & record** all information about the job, operator, tools, material, machine, conditions, drawing, layout.
3. **Divide the job into elements** (breaking up) and describe them with start / end break-points.
4. **Measure the time** taken by the operator for each element with the stop-watch; take a sufficient number of cycles (N).
5. **Assess the effective speed** of working of the operator relative to the standard (rating).
6. **Convert the observed times to normal (basic) times** : NT = OT × (Rating / 100).
7. **Determine the allowances** (relaxation, contingency, special) and add them.
8. **Compute the standard time** of the job = normal time + allowances.

5. Breaking the Job into Elements

- **Element** = a distinct part of a specified job, selected for convenience of observation, measurement and analysis. A job is divided so that: (i) the rating can be applied per element; (ii) elements with different characteristics are separated; (iii) the standard data library can be built; (iv) abnormal / foreign elements can be excluded.

Rules / guidelines for breaking into elements

- Elements should be **short enough** to be timed accurately but not so short that readings are inaccurate (minimum ≈ 0.04 min (2.4 s) for a watch reader; 0.1 min to be safe).
- Separate **manual** elements from **machine** (power-fed) elements; separate **constant** elements (same time every cycle) from **variable** elements (time depends on size, weight, length etc.).
- **Regular** elements in every cycle; **irregular / occasional** elements (e.g. sharpening a tool) recorded separately and spread over the cycles.
- **Foreign (extraneous) elements** – unavoidable delay or not part of the job (talking, dropped tool) – are noted and excluded from the time.

- Choose break-points that are clear: a sound, a visual change (e.g. "part clicks into fixture") – the end of one element = start of the next.
- **Governing element** = in a cycle with simultaneous elements, the longest (e.g. a machine run) governs the time.

Type of element	Meaning	Example (turning a shaft)
Repetitive / regular	Occurs every cycle	Pick up bar, clamp in chuck
Occasional	Occurs at intervals	Re-sharpen tool each 50 pieces
Constant	Same time every cycle	Start the machine
Variable	Time changes with the work	Turn length varies with diameter
Manual / Machine	Done by worker / by machine power	Load (manual) / feed (machine)
Foreign	Not part of the job	Operator wipes his face

6. Recording Information (Time Study Form)

- Record on the **top of the observation sheet:** study number, date, name and number of the operator, department, observer; description of the operation, part name / number, drawing, material, machine & tool (type, number, speed, feed), work-place layout sketch, conditions (light, temperature), start & end time of the study. All details that affect time, so that a future person can reproduce the study.
- **Methods of timing** using the stop-watch:

(a) **Continuous timing** – the watch runs continuously from the start; time is noted at the end of each element (reading R); element time = difference of successive readings. Foreign elements shown; no time lost; used for short elements.

(b) **Repetitive / snap-back timing** – the hands are returned to zero at the end of each element; the reading = element time. Simple, but time of the snap-back is lost, errors for short elements.

(c) **Accumulative timing** – two watches mounted on the board connected by a lever: when one runs the other is stopped; the second is read while the first runs. Direct element times without subtraction.

Note: Use a sufficient number of readings (6–10 cycles to begin with); discard abnormal / foreign readings.

Elem.	Description	Cycle 1	2	3	4	5	Avg (OT)	R%	NT
A	Pick up part	0.20	0.22	0.19	0.20	0.21	0.204	110	0.224
B	Drill hole (machine)	1.50	1.52	1.50	1.49	1.50	1.502	100	1.502

Elem.	Description	Cycle 1	2	3	4	5	Avg (OT)	R%	NT
C	Remove, place in bin	0.30	0.28	0.32	0.30	0.30	0.300	90	0.270

Note: Sample record: OT = observed time (average) in min; R = rating; NT = OT × R/100. Rating is not applied to the machine-controlled element (B).

Number of cycles to be timed

- Depends on cycle time & accuracy; ILO / General Electric table: cycle 0.10 min → 200 cycles; 0.25 → 100; 0.50 → 60; 0.75 → 40; 1.0 → 30; 2.0 → 20; 5.0 → 10; 10 → 8; 20 → 5; 40 → 3; over 40 → 2.
- Statistical formula (95 % confidence, ±5 % accuracy): $N = (40 \sqrt{(n \Sigma x^2 - (\Sigma x)^2) / \Sigma x)^2}$ n = number of readings in the trial, x = each reading.

7. Rating (Performance Rating / Levelling)

Definition

Rating is the assessment of the worker's rate of working relative to the observer's concept of the rate corresponding to **standard performance.**

Normal time = Observed time × (Rating / 100) (Rating factor R = observed rate / standard rate)

- Why rating? The operator observed may work faster or slower than the standard. Rating converts the observed time into the time a **normal, qualified worker** would take.

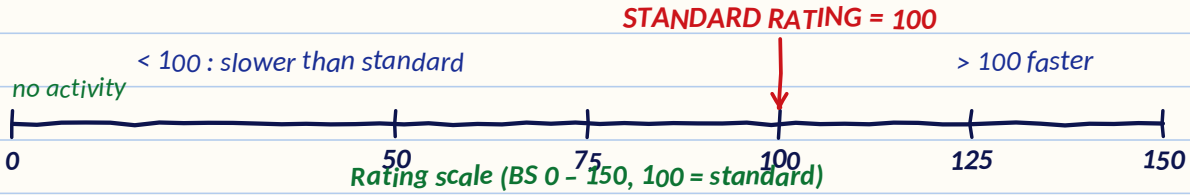
Standard performance and standard rating

- **Standard performance** (BS 3138): the rate of output which qualified workers will naturally achieve without over-exertion as an average over the working day, provided they know and adhere to the specified method and provided they are motivated to apply themselves to their work. It is denoted as 100 on the BS 0–100 scale.
- **Standard rating** = the rating corresponding to the standard performance (100 on the BS scale; 60 on the 60–80 scale; 75 on the 75–100 scale).
- Bench-marks: walking 4 miles / hour (≈ 6.4 km/h) without load on level ground = standard (100); dealing a pack of 52 playing cards into four piles in 0.5 minute = standard.

Scales of rating

Scale	Standard (normal)	Incentive (≈ +33 %)	Remarks
0 – 100 (British Standard)	100	133	0 = no activity; 100 = standard; > 100 faster
60 – 80 (American)	60	80	80 = incentive pace
75 – 100	75	100	100 = incentive pace
100 – 133	100	133	Same as BS scale

Note: Example: observed rating 90 on the BS scale = 90 % of standard speed → R = 0.90.



Systems of rating

1. **Speed rating** (pace rating): observer assesses only the speed / tempo of movement against the standard concept (single factor) – R = observed pace / standard pace. Simple, quick; depends on observer's judgement and training (use training films).
2. **Westinghouse system (levelling):** four factors – **Skill, Effort, Conditions, Consistency;** each graded (A to F) and a numeric ± value from the table; Rating factor = 1 + algebraic sum.
3. **Synthetic rating** (Morrow): the observed time of a few elements is compared with the PMTS (e.g. MTM) time; R = P / A (P = predetermined time, A = actual observed time); average R is applied to the other elements.
4. **Objective rating** (Mundel): two steps – (i) rate the speed against a standard reference (pace only); (ii) add a **job-difficulty adjustment** (amount of body used, foot pedals, bi-manual, eye-hand coordination, handling / sensory requirements, weight handled) from tables → Rating = Pace rating × Secondary adjustment factor.
5. **Performance rating (physical)** / skill & effort rating (Lowry et al.) – and "pace rating" with training films by the Society for Advancement of Management (SAM).

Westinghouse rating table

Grade	Skill	Effort	Conditions	Consistency
A1 / A	Super skill +0.15	Excessive +0.13	Ideal +0.06	Perfect +0.04
A2	+0.13	+0.12		
B1 / B	Excellent +0.11	Excellent +0.10	Excellent +0.04	Excellent +0.03

Grade	Skill	Effort	Conditions	Consistency
B2	+0.08	+0.08		
C1 / C	Good +0.06	Good +0.05	Good +0.02	Good +0.01
C2	+0.03	+0.02		
D	Average 0.00	Average 0.00	Average 0.00	Average 0.00
E1 / E	Fair –0.05	Fair –0.04	Fair –0.03	Fair –0.02
E2	–0.10	–0.08		
F1 / F	Poor –0.16	Poor –0.12	Poor –0.07	Poor –0.04
F2	–0.22	–0.17		

- **Skill** = proficiency in following a given method; **Effort** = willingness to work (speed with which skill is applied); **Conditions** = working conditions affecting the operator (not the operation) – light, temperature, noise; **Consistency** = variation in repeated times (fine, rating when elements repeat).
- Rating factor R = 1 + (Skill + Effort + Conditions + Consistency). Rating % = R × 100.

8. Allowances and Standard Time Determination

- Normal time is the time for a worker working at standard rating with **no allowance for rest, personal needs or delay.** A worker cannot work continuously; allowances are added to get a time which can be achieved in actual conditions.

Allowance	Purpose	Typical value
Personal needs (relaxation)	Drinking water, toilet, washing, tea; depends on climate and work	5 % (men), 7 % (women)
Basic fatigue	Energy consumed + monotony while working under ideal conditions	4 % of normal time
Variable fatigue	Special: standing / awkward posture, heavy force, poor light, heat, noise, close attention, monotony, mental strain	2 – 20 % (table / ILO)
Contingency	Small unavoidable delays, irregular elements not timed, e.g. tool change, inspection	≤ 5 %
Delay (unavoidable)	Machine breakdown, power failure, waiting for materials / instructions	by study / sampling
Special / policy	Training, setting-up, interference (multi-machine), tool & process allowances, learning, management policy	by policy

Standard time determination

Observed time (OT) → **Normal time NT = OT × R / 100** → add allowances → **Standard time ST = NT + (Allowances) = NT × (1 + A)** (A = allowance as a fraction of normal time).

When allowance is given as a % of the **total (shift) time** : **ST = NT / (1 – A)** (e.g. 15 % → ST = NT / 0.85).

Standard output per day = shift time / ST; Performance index (efficiency) = (standard time × output) / (time on the job) × 100.

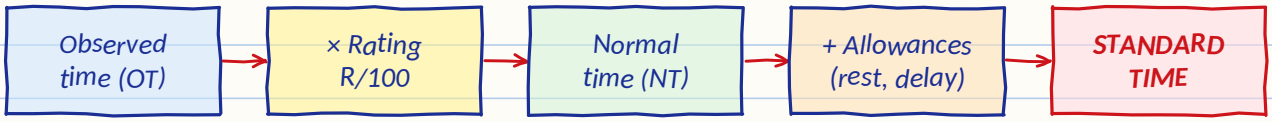


Fig 3.3 Build-up of standard time

9. Predetermined Motion Time Systems (PMTS)

- Definition: a work-measurement technique in which **times established for basic human motions** (classified by the nature of the motion and the conditions under which it is made) are used to build up the time for a job at the defined level of performance.
- Principle: break the job into basic motions (reach, move, turn, grasp, position, disengage, release, eye travel, body motions); look up the time for each from tables (depends on distance, weight, case, difficulty); add them. No stop-watch or rating needed, so it can be applied **before the job exists** (during the design of the method).

System	Developer / year	Time unit	Remarks
Work-Factor (WF)	Quick, Duncan & Malcolm (1938)	1 RWF = 0.0001 min	Work factors for body member, distance, manual control, weight
Methods-Time Measurement (MTM-1)	Maynard, Stegemerten & Schwab (1948)	1 TMU = 0.00001 h = 0.0006 min = 0.036 s	Most widely used; MTM-2, MTM-3 (faster, less detailed)
MOST (Maynard Operation Sequence Tech.)	Zandin (1967)	TMU (index × 10)	Sequence models: General move (ABGABPA), Controlled move, Tool use

System	Developer / year	Time unit	Remarks
BMT (Basic Motion Time Study)	Brisch / Mackie (1950s)	1 BMU = 0.0001 min	Basic motions with time tables
DMT / Dimensional Motion Times	Wygant & Sellie	—	Dimension-based motion times

- MTM-1 basic motions: **Reach (R), Move (M), Turn (T), Apply pressure (AP), Grasp (G), Position (P), Release (RL), Disengage (D),** Eye travel (ET) & Eye focus (EF), body, leg and foot motions (walk, bend, kneel, sit).
- Conversions (MTM): 1 TMU = 0.036 s; 1 min = 1666.7 TMU; 1 h = 100 000 TMU.

Merits of PMTS	Demerits of PMTS
No stop watch or rating – objective & consistent	Needs trained analyst; data tables are costly (licence)
Method can be assessed before production starts	Detailed analysis takes long time; not for long-cycle / mental work
Helps in method improvement, plant layout, line balancing	Not accepted by some workers; applicable to manual work only
Gives accurate standard data library	Limited to the motions in the tables

10. Factors Affecting the Rate of Working

- Rate of working (speed) of an operator is not constant; it depends on:

Factor	Examples
Operator	Skill, training, experience, health, age, physical fitness, attitude, motivation, mood, personal problems
Job / method	Difficulty, accuracy needed, weight, work-place layout, tools & fixtures, motion economy of the method
Material / machine	Quality of material, machine condition, speeds & feeds, tooling
Environment	Temperature, humidity, lighting, noise, ventilation, vibration, cleanliness
Organisation / management	Supervision, incentive scheme, wage system, job security, group relations, work load, time of day, shift, fatigue pattern
Time-study situation	Awareness of being observed, observer's attitude, union stand, fear of rate cutting

Note: Management must provide an environment where the worker can attain standard performance without excessive fatigue.

11. Standard Data, Merits & Limitations of Time Study

- **Standard data** (synthesis): element times (constant & variable) collected from many earlier time studies, tabulated or as formulas / curves (e.g. time = a + b × length). New jobs are built up by adding the standard element times – no new stop-watch study; consistent standards, quick estimates, lower cost.
- Example: drilling time T = (L / (f × N)) where L = depth + approach (mm), f = feed (mm/rev), N = rpm; handling time from standard data tables per weight / distance.

Merits of time study	Limitations of time study
Gives direct, observed, and accurate time for short repetitive work	Needs trained observers; rating is subjective (±10 % error)
Basis for fair standards, planning, cost, incentive wages	Not suited to long-cycle, non-repetitive & mental work
Detects delays & method faults during the study	Workers may feel nervous / act abnormally when watched
Simple equipment; widely accepted by industry	Cost & time of repeated studies; becomes obsolete when method changes

Point	Time study	Work sampling
Observation	Continuous, with stop watch	Random instantaneous glimpses
Job type	Short, repetitive, manual	Long, irregular, group work
Rating	Needed	Needed only if standard time is desired
Output	Element time, standard time	% of time in each activity, delays, allowances
Operator effect	Observed continuously – may be disturbed	Less disturbing

Precautions while rating and timing

- Rate each element **while it is being performed** (not after); rate only the speed of working, not the operator's skill unless the system requires it; ignore the effect of observer's personal liking.
- Do not rate the machine-controlled elements; use training films / many practice sessions to calibrate the rating (check against other observers: rating accuracy ±5 %).

- Use the **average of the selected readings** after rejecting abnormal values (> 2 × the usual); average the ratings of the elements with different rating factors separately.

Learning curve and fatigue (side notes)

- New operators need time to reach standard output – training allowance / learning curve (the time per unit falls by a constant % each time output is doubled: 80 % curve is typical).
- Fatigue is reduced by good method (motion economy), proper rest breaks, good environment, rotation of jobs; the relaxation allowance recovers the energy expended.

12. Problems on Time Study, Rating and Allowances

P1. The average observed time for an element is 0.50 min. The worker is rated 110 %. Allowances are 15 % of normal time. Find the standard time and the output for an 8-hour shift.

Sol: NT = 0.50 × 1.10 = 0.55 min. ST = 0.55 × (1 + 0.15) = **0.6325 min.** Output = 480 / 0.6325 = **759 pieces / shift** (≈ 758.8).

P2. Same data, but the 15 % allowance is given as % of the total (shift) time. Find the standard time.

Sol: ST = NT / (1 – 0.15) = 0.55 / 0.85 = **0.647 min.**

P3. A time study gave average times & ratings: (1) pick-up 0.20 min, 110 %; (2) machining (machine-controlled) 1.50 min, 100 %; (3) remove & place 0.30 min, 90 %; (4) inspect 0.25 min, 120 %. Allowances 15 %. Find the standard time and output per 8 h.

Sol: NT = 0.20×1.10 + 1.50×1.00 + 0.30×0.90 + 0.25×1.20 = 0.22 + 1.50 + 0.27 + 0.30 = 2.29 min. ST = 2.29 × 1.15 = **2.63 min.** Output = 480 / 2.63 = **182 pieces.**

P4. Westinghouse rating: skill – good (C1), effort – good (C2), conditions – average (D), consistency – excellent (B). Observed time 0.90 min. Find the normal time.

Sol: Skill +0.06; Effort +0.02; Conditions 0; Consistency +0.03 → sum = +0.11. R = 1.11. NT = 0.90 × 1.11 = **0.999 ≈ 1.0 min.**

P5. Ten stop-watch readings (min): 0.5, 0.6, 0.5, 0.7, 0.6, 0.5, 0.6, 0.5, 0.7, 0.6. Find the number of cycles to be timed (95 % confidence, ±5 %).

Sol: $\Sigma x = 5.8$; $\Sigma x^2 = 3.42$; $n = 10 \rightarrow n\Sigma x^2 - (\Sigma x)^2 = 34.2 - 33.64 = 0.56$; $\sqrt{0.56} = 0.748$. $N = (40 \times 0.748 / 5.8)^2 = (5.16)^2 = 26.6 \approx$ **27 cycles.**

P6. Observed average cycle 2.0 min; rating 85 %; allowances: personal 5 %, fatigue 5 %, delay 3 %. Find standard time.

Sol: NT = 2.0 × 0.85 = 1.70 min; total allowance = 13 %; ST = 1.70 × 1.13 = **1.92 min.**

P7. A job analysed by MTM needs 250 TMU. Find the time in seconds and minutes; with 10 % allowance find standard time.

Sol: Time = 250 × 0.036 = 9.0 s = 0.15 min. ST = 9.0 × 1.10 = **9.9 s = 0.165 min.**

P8. An operator makes 300 pieces in 8 h; standard time of the job = 1.4 min. Find the performance index.

Sol: Standard minutes produced = 300 × 1.4 = 420; available time = 480 min.

Performance = 420 / 480 × 100 = **87.5 %.**

P9. An operator attends 3 identical machines. Loading / unloading time per machine = 1.2 min, machine running time = 4.0 min, walking time between machines = 0.2 min. Find the machines per operator (ideal) and the cycle time for 3 machines.

Sol: $n = (l + m)/(l + w) = (1.2 + 4.0)/(1.2 + 0.2) = 5.2 / 1.4 = 3.7$ → check $n = 3$ and $n = 4$.

For $n = 3$: machine-cycle = $l + m = 5.2$ min; operator work = $3 \times (1.2 + 0.2) = 4.2$ min < 5.2 → cycle time = **5.2 min per 3 pieces** (machine-governed); operator idle = 1.0 min per cycle.

P10. A job has a normal time of 8 min. Allowances: personal 5 %, fatigue 4 %, special 3 %, all as % of the total (shift) time. Find ST and the output in 8 h.

Sol: total allowance = 12 % → ST = 8 / (1 – 0.12) = **9.09 min.** Output = 480 / 9.09 = **52.8 ≈ 52 pieces / shift.**

13. Quick Revision & Probable Questions

Memory hooks

- ST = (OT × R/100) × (1 + A); with shift-based allowance: ST = NT/(1 – A).
- Westinghouse: Skill, Effort, Conditions, Consistency (S E C C).
- Timing: continuous – snap-back – accumulative. Wink / TMU: 1 TMU = 0.036 s.
- Time study steps: Select – Record – Break – Measure – Rate – Normal – Allowances – Standard.

1. Define time study. Explain the equipment used and the steps in a time study.
2. How is a job broken into elements? State the rules. Explain methods of timing.
3. What is rating? Explain standard performance and the scales of rating. Describe the Westinghouse system.
4. List the allowances. How is standard time determined?
5. What is PMTS? Explain MTM / Work Factor / MOST with merits and demerits.
6. Discuss the factors affecting the rate of working.
7. Numericals on rating, allowances, number of cycles and standard time (P1 – P8).